ASX Factor Tilt Portfolio, Monthly Rebalance Backtest

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February 2026

Abstract

This report documents a simple long only, monthly rebalanced equity strategy applied to a large cap Australian equity universe, namely the ASX50 constituents as of the 2nd of January 2026. The strategy uses two cross sectional, price derived signals, a nine month momentum signal and a defensive proxy based on semi annual inverse volatility, combined into a weighted composite score with a heavy defensive tilt (0.9 defensive, 0.1 momentum). Each month, the top ten names by composite score are held in equal weights for one month.

Over the current backtest sample (month end observations from January 2021 to December 2025, 60 monthly observations), the strategy delivered a cumulative return of 52.79%, a CAGR of 8.85%, annualised volatility of 11.39%, Sharpe ratio of 0.804 and a maximum drawdown of -10.25%. Over the same sample, benchmark ETF proxies A200.AX and SFY.AX delivered lower cumulative returns and lower Sharpe ratios, with deeper drawdowns.

Overall, the gross results suggest that a defensive tilted, monthly rebalanced selection rule can improve risk adjusted outcomes versus broad equity ETFs over this sample, but the conclusion is conditional on survivorship and data adjustment assumptions, realistic execution timing and net of cost performance after applying transaction costs consistent with the observed turnover.

1 Introduction

This report documents a simple, long only, monthly rebalanced equity strategy on an ASX large cap universe. Stocks are scored cross sectionally using two price derived signals, a momentum proxy (nine month price change) and a defensive proxy (lower recent volatility, implemented as six month inverse volatility). Each month, the portfolio holds the top ten stocks by a weighted composite score, equal weighted, then rebalances at the next month end. Performance is compared to standard Australian equity benchmark ETF buy-and-hold strategies .

The objective is not to claim a production ready investment process but to test whether a defensive tilt combined with a modest momentum overlay can improve return per unit of risk relative to broad market Australian equity exposures over the sample period.

2 Research question and hypothesis

Does a monthly rebalanced, top ten ASX large cap portfolio selected using a composite of nine month momentum and inverse six month volatility produce higher risk adjusted performance, measured by Sharpe

ratio and drawdown than investable Australian equity benchmark ETFs over January 2021 to December 2025?

The hypotheses tested are as follows. First, the composite score strategy should maintain a similar Sharpe ratio to the A200.AX and SFY.AX ETFs over the sample period. Second, the strategy will deliver a shallower maximum drawdown than the benchmarks due to the heavy tilt on the inverse volatility (defense/quality proxy) factor. Third, the strategy will exhibit non trivial turnover under monthly rebalancing, so net performance is sensitive to transaction cost assumptions and will be lower than gross performance.

3 Data

3.1 Universe

The investable universe is based on the S&P/ASX 50 constituent list, scraped from the public Wikipedia page for the index and then converted to Yahoo Finance ticker format by appending the suffix ".AX" to each symbol.

In the notebook workflow, the constituent symbols are first scraped from Wikipedia, then transformed into Yahoo Finance style tickers (for example, `CBA` becomes `CBA.AX`).

3.2 Prices and returns

The strategy uses daily price data (adjusted close) that is then resampled to month end prices. In the current workflow, prices are loaded from a local CSV file (`ASX50.csv`). Monthly prices are defined as the last available observation in each month:

$$P_{i,t}^{(M)} = \text{month end close for stock } i \text{ in month } t.$$

Monthly simple returns are then computed as:

$$R_{i,t} = \frac{P_{i,t}^{(M)}}{P_{i,t-1}^{(M)}} - 1.$$

The notebook also drops columns with missing values after monthly resampling in the strategy universe preprocessing step.

3.3 Benchmarks

Two Australian equity ETF proxies are used as benchmarks:

- `A200.AX`
- `SFY.AX`

A200.AX and SFY.AX are used because they are liquid, investable ETF proxies for broad Australian equity exposure, making them practical comparators for a long only ASX large cap strategy. They provide simple buy and hold alternatives with low operational complexity, which helps interpret whether any observed outperformance is large enough to justify the added turnover and implementation complexity of the active selection rule. Moreover SFY.AX possesses a similar stock universe to the strategy's universe, namely the ASX50.

4 Methodology

4.1 Rebalance schedule and information set

The strategy is implemented at a monthly frequency using month end formation dates. At each month end t, signals are computed from historical price data available up to that month end. Stocks are then ranked cross sectionally, the top ten are selected and portfolio return is measured over the next month.

The forward one month return is defined in the notebook as:

$$R_{i,t}^{\text{fwd}} = \frac{P_{i,t+1}^{(M)}}{P_{i,t}^{(M)}} - 1.$$

This means the return value indexed by date t is the return earned from month end t to month end $t + 1$. In other words, the return is *labeled by the formation date*.

Timing note: The current implementation is consistent with a month end formation, month ahead holding convention, but it implicitly assumes rebalancing at the same month end close used to compute signals.

4.2 Signal definitions

4.2.1 Momentum proxy

The momentum signal is a nine month price change, measured from month end closes:

$$m_{i,t} = \frac{P_{i,t}^{(M)}}{P_{i,t-9}^{(M)}} - 1.$$

This is a simple price momentum proxy and does not currently exclude the most recent month return. It is therefore a straightforward nine month lookback signal rather than a standard 12 month minus 1 month institutional momentum specification.

A nine month lookback is used as a simple, transparent momentum proxy that fits cleanly within the five year sample while retaining enough observations for monthly ranking and portfolio formation. This choice is exploratory and is treated as a specification to be stress tested against alternative momentum definitions such as 6 month, 12 month and 12 month minus 1 month momentum.

4.2.2 Defensive proxy via volatility

The defensive proxy is based on recent volatility using a rolling six month standard deviation of monthly returns:

$$\sigma_{i,t}^{(6)} = \mathrm{sd}(R_{i,t-5}, \ldots, R_{i,t}).$$

The raw defensive signal is inverse volatility:

$$v_{i,t} = \frac{1}{\sigma_{i,t}^{(6)}}.$$

Higher $v_{i,t}$ corresponds to lower recent volatility, which is treated as more desirable in the composite score.

4.3 Cross sectional scoring

Raw signals are converted into cross sectional ranks each month, with higher raw signal values receiving better ranks. In the notebook, the ranking logic is implemented using descending ranks by column after reshaping the panel so that each date is ranked across the available names.

The rank to score mapping is:

$$s_{i,t} = 1 - \frac{\mathrm{rank}_{i,t} - 0.5}{N},$$

where N is the number of eligible stocks in the cross section used by the scoring function for that run.

This transformation maps better ranked names to higher scores and places scores approximately in the interval $(0, 1)$, with the top ranked stock receiving a score close to 1 and the bottom ranked stock receiving a score close to 0. Cross-sectional N was found to be 45, as some stock pricing data were unavailable for the full length of the testing timeframe.

4.4 Composite score

The composite stock score is a weighted combination of momentum and defensive scores:

$$\mathrm{Score}_{i,t}^{\mathrm{agg}} = 0.1 \cdot s_{i,t}^{\mathrm{mom}} + 0.9 \cdot s_{i,t}^{\mathrm{def}}.$$

The 0.9 weight on the defensive proxy creates a strong low volatility tilt, while the 0.1 momentum weight adds a smaller trend component.

The 0.9 and 0.1 weighting is an intentional defensive tilt, designed to prioritise lower volatility names while still allowing the momentum signal to influence selection at the margin. The weights were chosen as a baseline specification rather than as an optimised parameter set and hence should be further evaluated with a sensitivity analysis by testing alternative weight splits.

4.5 Portfolio construction

At each month end formation date t:

1. Compute the aggregate score for each eligible stock.
2. Rank stocks cross sectionally by aggregate score.
3. Select the top $K = 10$ names.
4. Assign equal weights of $1/K$ to selected names.
5. Hold the portfolio for one month.

In the notebook implementation, the selected set is applied by retaining forward returns for names with rank ≤ 10 and replacing all other names with zero, then averaging across the ten selected names.

Formally, if $I_{i,t}$ indicates whether stock i is selected at time t,

$$I_{i,t} = \begin{cases} 1, & \text{if } \operatorname{rank}_{i,t}^{\text{agg}} \leq 10, \\ 0, & \text{otherwise,} \end{cases}$$

then the portfolio return is:

$$R_t^p = \frac{1}{10} \sum_i I_{i,t} \, R_{i,t}^{\text{fwd}}.$$

4.6 Backtest return calculation

The current sample of reported results runs from month end January 2021 to month end December 2025, which corresponds to 60 monthly forward labeled portfolio returns in the notebook output.

The cumulative equity curve is built as growth of \$1:

$$\text{Equity}_t = \prod_{\tau \leq t} (1 + R_\tau^p),$$

and cumulative return is:

$$\text{CumRet}_t = \text{Equity}_t - 1.$$

Labeling convention note: Because forward returns are indexed by formation date, the row labeled 2021-01-31 corresponds to the return earned from 2021-01-31 to 2021-02-28 (i.e. the portfolio performance as a result of 2021-01-31's buying decisions).

5 Transaction costs and turnover

A simple proportional transaction cost deduction has been applied to the strategy returns. Turnover is computed and reported as an implementation diagnostic for a monthly rebalanced top ten portfolio.

Current turnover summary for the strategy:

- Average monthly turnover: 0.2816666667 (28.17%)
- Annualized turnover: 3.3800000000 (338.00%)
- Maximum monthly turnover: 0.7000000000 (70.00%)

These turnover levels indicate that transaction cost assumptions can materially affect realised performance, especially if the strategy is applied to names with wider spreads or lower liquidity.

Turnover is computed from changes in the target portfolio weights across rebalancing dates. Let $w_{i,t}$ denote the target weight for stock i at formation date t, where $w_{i,t} = 1/10$ for selected names and $w_{i,t} = 0$ otherwise. One way turnover at month t is computed as

$$\mathrm{TO}_t = \frac{1}{2} \sum_i \left| w_{i,t} - w_{i,t-1} \right| .$$

This is implemented directly from the month to month differences in the target equal weight selection vector and it does not attempt to model intra month weight drift from asset returns.

A proportional transaction cost model is applied using one way turnover. Let c denote the one way cost per dollar traded (in decimal form). In this backtest, $c = 0.001$ (10 bps per side). The monthly cost drag is modelled as $2c \cdot \mathrm{TO}_t$, where the factor of two represents buys and sells required to rebalance. Net portfolio return is computed as

$$R_t^{p,\mathrm{net}} = R_t^p - 2c \cdot \mathrm{TO}_t .$$

The cost parameter should be stress tested across plausible values.

6 Performance evaluation

6.1 Primary metrics

The main performance metrics used in this report are:

- Cumulative return
- CAGR (compound annual growth rate)
- Annualised volatility (standard deviation of monthly returns, annualised by $\sqrt{12}$)
- Sharpe ratio (annualised from monthly excess returns)
- Maximum drawdown

6.2 Secondary diagnostics

In addition to the primary performance metrics, this report tracks turnover because it is directly relevant to implementation feasibility and net of cost performance. A useful additional extension is to report Sortino ratio,

win rate, best month, worst month and drawdown duration once those figures are finalised. The following statistics are based off of the backtest period between 2021-01-31 to 2025-12-31.

Table 1: Performance summary, gross of transaction costs, current notebook output

Series	Cum. Ret. (%)	CAGR (%)	Vol. (%)	Sharpe	Max DD (%)	Turnover
Strategy	52.79	8.85	11.39	0.804	-10.25	3.38x p.a.
A200.AX	31.23	5.59	11.57	0.528	-15.17	N/A
SFY.AX	27.57	4.99	11.99	0.467	-14.81	N/A

Table 2: Additional diagnostics currently available in the notebook output

Series	Peak Cum. Ret. (%)	Final wealth of $10,000	Peak wealth of $10,000	Max monthly turnover
Strategy	64.10	$15,279	$16,410	70.00%
A200.AX	35.78	$13,123	$13,578	N/A
SFY.AX	33.99	$12,757	$13,399	N/A

Interpretation of "Peak Cum. Ret.": The provided values labeled “Peak Cum. Ret. (%)” are larger than the final cumulative returns and are therefore interpreted here as the *peak cumulative return reached during the sample*, not the best one month return.

7 Results

7.1 Headline results

On the current gross basis and over the 2021 to 2025 sample, the strategy outperformed both benchmark ETFs on cumulative return, CAGR and Sharpe ratio, while also exhibiting slightly lower annualised volatility and materially lower maximum drawdown.

Relative to A200.AX, the strategy delivered:

- +21.55 percentage points higher cumulative return (52.79% vs 31.23%)
- +3.26 percentage points higher CAGR (8.85% vs 5.59%)
- +0.276 higher Sharpe ratio (0.804 vs 0.528)
- 0.18 percentage points lower annualised volatility (11.39% vs 11.57%)
- 4.93 percentage points shallower maximum drawdown (10.25% vs 15.17% in absolute drawdown magnitude)

Relative to SFY.AX, the strategy delivered:

- +25.21 percentage points higher cumulative return (52.79% vs 27.57%)
- +3.86 percentage points higher CAGR (8.85% vs 4.99%)

- +0.337 higher Sharpe ratio (0.804 vs 0.467)
- 0.60 percentage points lower annualised volatility (11.39% vs 11.99%)
- 4.56 percentage points shallower maximum drawdown (10.25% vs 14.81% in absolute drawdown magnitude)

These net results are encouraging from a risk adjusted perspective, but they should be interpreted alongside the observed turnover (338% annualised), because implementation frictions may reduce net outperformance due to varying transaction costs.

7.2 Equity curve

Figure 1 plots the growth of $10,000 for the strategy and both benchmarks over the common sample window.

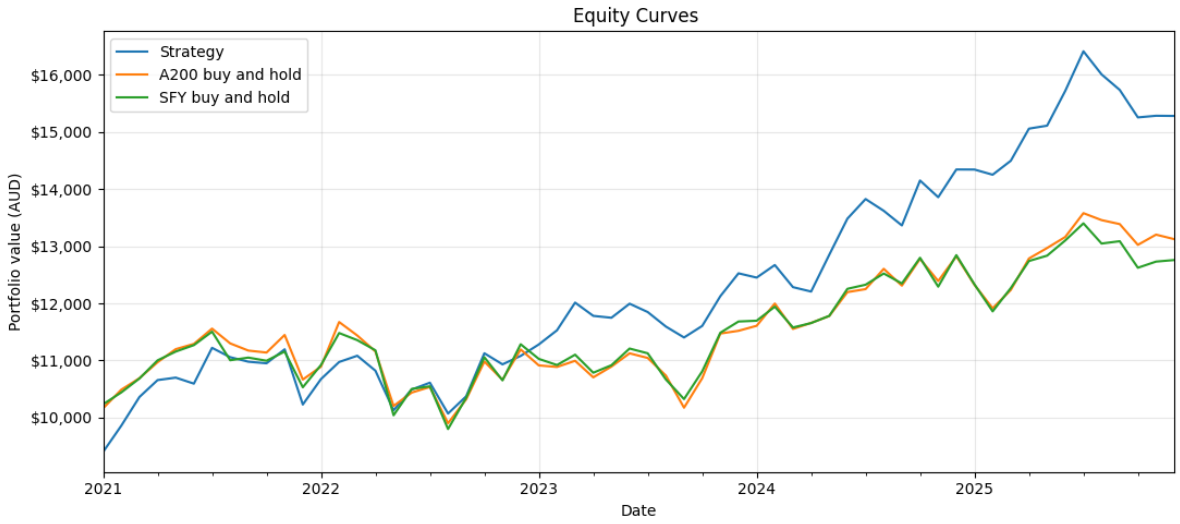

Figure 1: Growth of $10,000, strategy vs benchmarks

The equity curve shows steady compounding over the sample with intermittent drawdowns consistent with a concentrated top ten portfolio. The strategy reaches a peak cumulative return of 64.10% during the sample before finishing at 52.79%, indicating that the terminal value is below the prior peak. The largest equity decline occurs over the late 2021 to mid 2022 window, after which the strategy recovers and resumes an upward trajectory. Relative to the benchmark ETFs, the strategy achieves a higher terminal wealth while exhibiting a shallower maximum drawdown, consistent with the heavy inverse volatility tilt.

7.3 Drawdowns

The strategy maximum drawdown is -10.2493%, which is materially shallower than the benchmark drawdowns in the current sample (A200.AX at -15.1750%, SFY.AX at -14.8115%).

This is consistent with the defensive weighting in the composite score, where 90% of the ranking score is tied to inverse volatility. A strong defensive tilt should generally reduce downside sensitivity in risk off periods,

though this relationship is regime dependent and should be tested across longer samples.

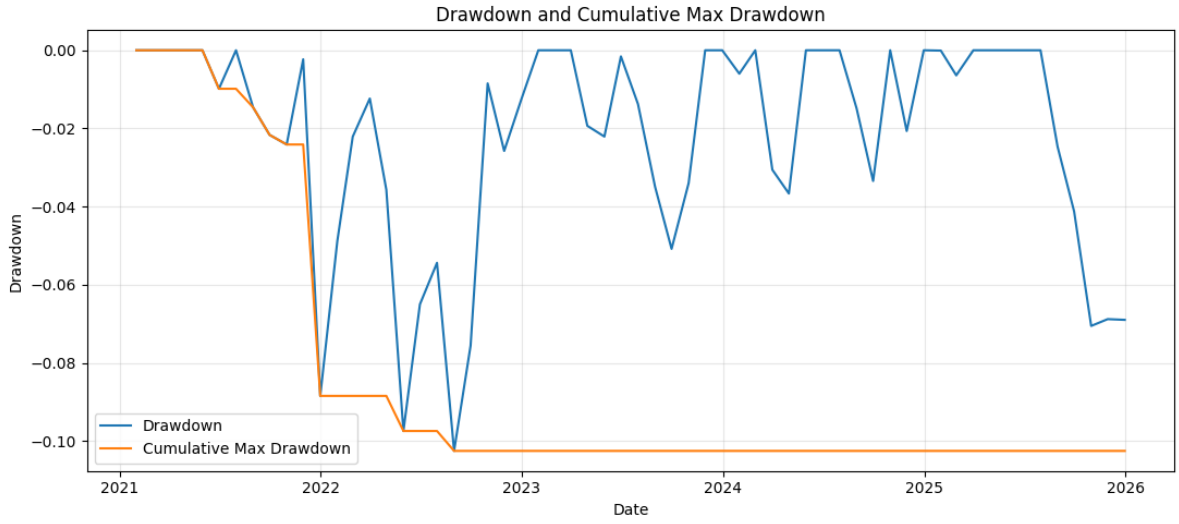

Figure 2: Drawdown series, strategy vs benchmarks

For the strategy, the worst drawdown occurs at the month end 2022-08-31, when the drawdown reaches -10.25%. The episode begins after a local peak at 2021-11-30 (drawdown at 0.0), followed by an initial sharp decline in December 2021 to about -8.84% at 2021-12-31, then a further leg down into mid 2022, with the drawdown deepening again to roughly -9.74% by 2022-05-31 and ultimately to the trough in August 2022. Overall, this is best characterised as a multi month drawdown with one sharp selloff month early in the episode and a slower grind lower thereafter as opposed to a single isolated shock month.

8 Robustness checks

The current results are based on one specific specification, nine month momentum, six month volatility, 0.9 and 0.1 composite weights, top ten equal weight selection, monthly rebalance and gross returns. For further improvements, following robustness checks are to be carried out before drawing strong conclusions:

1. **Alternative momentum window:** Compare 9 month momentum to 6 month, 12 month and 12 minus 1 month momentum.
2. **Alternative volatility window:** Test 3 month, 6 month and 12 month rolling volatility.
3. **Selection breadth:** Compare top 5, top 10 and top 15 selections.
4. **Composite Scoring/Weights:** Test 100% defensive, 75 and 25, 50 and 50 and 100% momentum and other scoring variations.
5. **Transaction cost sensitivity:** Apply multiple one way cost assumptions, for example 30 bps and 60 bps.
6. **Subperiod analysis:** Split 2021 to 2025 into smaller regimes to test stability of outperformance.

7. **Universe and survivorship controls:** Reconstruct historical constituent membership if possible, or at minimum discuss the impact of using a current constituent list.

9 Limitations and risks

Several limitations apply to the current prototype and are acknowledged as follows:

- **Survivorship bias risk:** If the universe is based on a current ASX 50 constituent list rather than historical membership by date, historical results may be biased upward.
- **Execution timing simplification:** The current design forms signals and applies them at the same month end close used in the signal calculation.
- **Transaction costs not yet deducted:** Annualized turnover of 338% suggests net returns may be meaningfully lower than gross returns.
- **Concentration risk:** A top ten equal weight portfolio can be materially concentrated in sectors or styles, even within a large cap universe.
- **Regime dependence:** A defensive and momentum mix may perform differently across inflation shocks, liquidity regimes and strong cyclical rotations.
- **Signal scope:** Signals are price derived proxies only and do not include fundamentals, valuation, or quality accounting data.

10 Conclusion

The current gross backtest indicates that a defensive tilted, momentum augmented top ten ASX large cap strategy outperformed two broad Australian equity ETF benchmarks on both absolute and risk adjusted metrics over the 2021 to 2025 sample, while also exhibiting lower drawdowns. The strategy produced a higher CAGR and Sharpe ratio than A200.AX and SFY.AX with similar or slightly lower annualised volatility.

At the same time, the strategy exhibits substantial turnover and the backtest still requires clear documentation of data adjustments and survivorship controls.

In summary, the backtest indicates that a defensive tilted composite score strategy, combining inverse six month volatility with smaller nine month momentum delivered higher gross cumulative return and a higher Sharpe ratio than A200.AX and SFY.AX over January 2021 to December 2025 while also exhibiting a shallower maximum drawdown. However, turnover is substantial under monthly rebalancing, so net performance is likely sensitive to transaction costs and execution assumptions. Further research is required to find cheap solutions to mitigate survivorship bias, followed by robustness tests across windows, weights and portfolio breadth. A more ambitious outlook would involve incorporating fundamental data into the cross sectional scoring process to account for value factors and quality factors such as debt-to-equity or earnings variability.

A Definitions

This appendix collects the core mathematical definitions used in the current strategy implementation.

Monthly return

$$R_{i,t} = \frac{P_{i,t}^{(M)}}{P_{i,t-1}^{(M)}} - 1$$

Nine month momentum

$$m_{i,t} = \frac{P_{i,t}^{(M)}}{P_{i,t-9}^{(M)}} - 1$$

Six month inverse volatility

$$\sigma_{i,t}^{(6)} = \mathrm{sd}(R_{i,t-5}, \ldots, R_{i,t}), \qquad v_{i,t} = \frac{1}{\sigma_{i,t}^{(6)}}$$

Rank to score mapping

$$s_{i,t} = 1 - \frac{\mathrm{rank}_{i,t} - 0.5}{N}$$

Aggregate score

$$\mathrm{Score}_{i,t}^{\mathrm{agg}} = 0.1 \cdot s_{i,t}^{\mathrm{mom}} + 0.9 \cdot s_{i,t}^{\mathrm{def}}$$

Forward one month return (formation date labeled)

$$R_{i,t}^{\mathrm{fwd}} = \frac{P_{i,t+1}^{(M)}}{P_{i,t}^{(M)}} - 1$$

Portfolio return

$$R_t^p = \frac{1}{10} \sum_i I_{i,t} \, R_{i,t}^{\mathrm{fwd}}$$

Equity curve and cumulative return

$$\mathrm{Equity}_t = \prod_{\tau \le t} (1 + R_\tau^p), \qquad \mathrm{CumRet}_t = \mathrm{Equity}_t - 1$$

Maximum drawdown

$$\mathrm{DD}_t = \frac{\mathrm{Equity}_t}{\max_{\tau \le t} \mathrm{Equity}_\tau} - 1, \qquad \mathrm{MDD} = \min_t \mathrm{DD}_t$$

Exact metrics used in this draft (current notebook output)

Metric	Value
Strategy cumulative return (%)	52.78689051318488
Strategy annualised volatility (%)	11.391578899006719
Strategy peak cumulative return (%)	64.10491139837804
Strategy maximum drawdown (%)	-10.249268657255751
Strategy CAGR (%)	8.84718917075411
Strategy Sharpe ratio	0.8038501113373321
Strategy average monthly turnover	0.2816666666666666
Strategy annualised turnover	3.3799999999999994
Strategy maximum monthly turnover	0.7000000000000002
A200.AX cumulative return (%)	31.233619521011956
A200.AX annualised volatility (%)	11.572037201361608
A200.AX peak cumulative return (%)	35.77948486217779
A200.AX maximum drawdown (%)	-15.174976772994754
A200.AX CAGR (%)	5.586652531052683
A200.AX Sharpe ratio	0.5279682108411473
SFY.AX cumulative return (%)	27.573843264469854
SFY.AX annualised volatility (%)	11.989096214461169
SFY.AX peak cumulative return (%)	33.993512036878926
SFY.AX maximum drawdown (%)	-14.811516770555066
SFY.AX CAGR (%)	4.991061792863993
SFY.AX Sharpe ratio	0.4665858001295411